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FACULTY OF ENGINEERING

INGB417 – GROUP ASSIGNMENT

Group 9

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1. Executive Summary

2. Background

3. Design Problem and Objectives

4. Design Requirements

5. Design Assumptions and Constraints

5.1. Process Flow

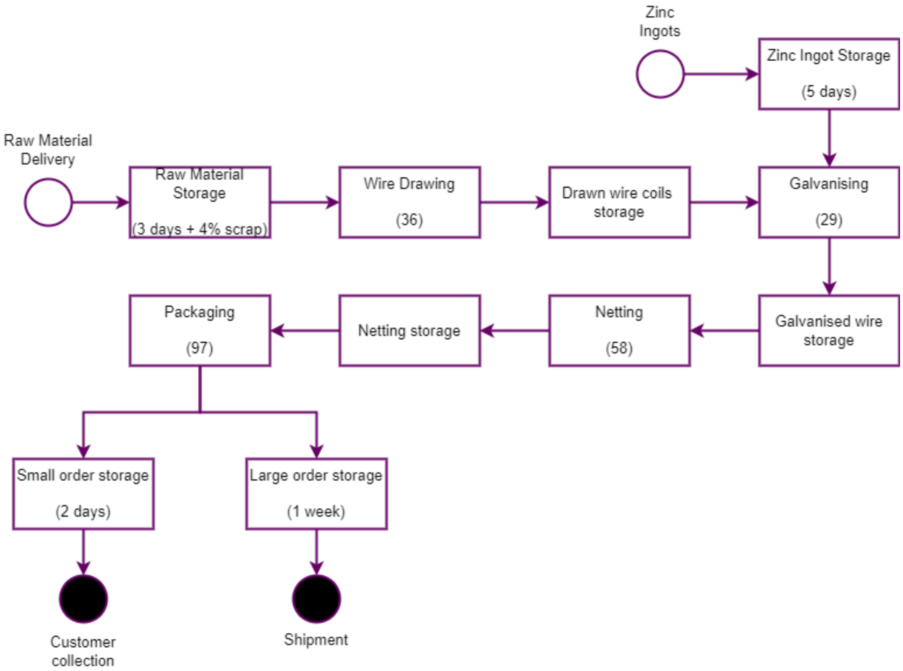

Figure 5.1.1: High-level Process Flow Diagram

5.2. Current State and Given Information

Operation A (Wire Drawing)

- Steel wire coils used
  - 5mm diameter
  - 7000m length
- Drawn wire rolled into coils before galvanization

Table 5.1.1: Wire diameters of drawn wire

Product	Wire diameter (mm)
Bird Netting	0.6
Chicken Netting	0.9
Jackal Netting	1.8
Pig Netting	1.8
Welded Mesh	2.5

Operation B (Galvanising of wire)

- Wire is drawn through bath with liquid
- Zinc ingots bought to keep process going

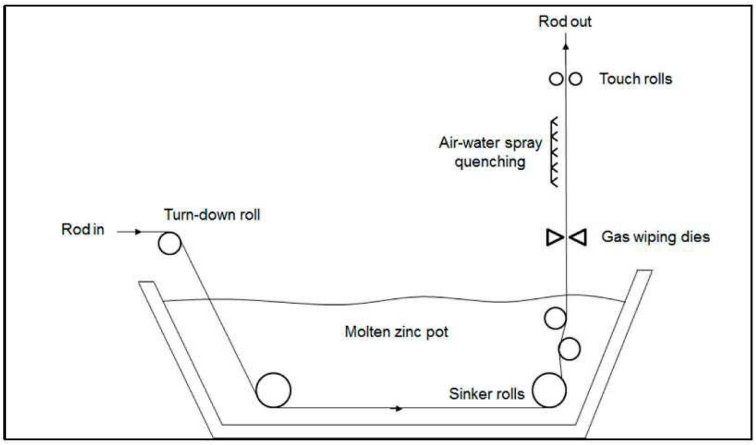

Figure 5.1.1: Concept Diagram of Galvanisation Process

Operation C (Netting Production)

- Total annual demand of 26 000 tons

Table 5.1.2: Product Specifications

	Roll Specifications					
Product	**Height (m)**	**Length (m)**	**Diameter (m)**	**Weight (kg)**	**Aperture (mm)**	**Annual Demand (tons)**
Bird Netting	0.9	50	0.25	22.5	13	4000
Chicken Netting	1.8	50	0.28	41.9	25	5500
Jackal Netting	0.6	50	0.37	19.5	75	2500
Pig Netting	0.9	30	0.29	14.7	90	8000
Welded Mesh	2	30	0.34	70	50	6000

Operation D (Packaging)

No specifications were given for packaging. Research done about this and decisions made about which machines to use can be seen in section 6.5 (Operation D (Packaging)) of this document.

6. Detail Design Specification

6.1. Raw Material Storage

The raw material of this process is steel wire coils of 7km long, weighing 2.5 tons and with a diameter of 5mm. It is assumed that the diameter of a coil is 1m.

A total of three days storage space is needed for production. 148 coils are needed to meet this demand. Coils are stored in racks of six coils per rack as can be seen in figure 6.1.1 below. These racks will have wheels with locks on them for easy movement of the racks when attached to vehicles and locked when needed to stay in place. The storage racks have dimensions of 4m*1m*1.2m (l*w*h), (Ross Industrial, 2024).

Figure 6.1.1: Raw Material Storage Racks

The total area needed for one rack (including walkway around it) is 18 m^2. A total of 25 racks will be needed to satisfy the storage requirements. The total area thus required for raw materials storage for three days is 450m^2.

6.2. Operation A (Drawing)

Specifications

Figure 6.2.1: Model 450 Drawing Machine

Figure 6.2.1 above shows drawing machine model 450, used to draw the wires from 5 mm to 1.8 mm for jackal netting and pig netting and to 2.5 mm for welded mesh. This same machine is also used to draw wires from 5 mm to 1.8 mm for bird netting and chicken netting before entering drawing machine model 300 to be drawn even thinner as drawing machine model

450 has a minimum diameter of 1 mm it can draw a wire. Figures 6.2.2 and 6.2.3 below show the drawing machine Model 300, used to draw the wires to 0.6 mm and 0.9 mm for bird netting and chicken netting.

Figure 6.2.2: Model 300 Drawing Machine Input

Figure 6.2.3: Model 300 Drawing Machine Output

Table 6.2.1: Drawing Machine Specifications

Model 300	Model 450
Max Diameter in: 2.8 mm	Max Diameter in: 5mm
Min Diameter out: 0.5 mm	Min Diameter out: 1 mm
Max speed: 25 m/s	Max speed: 16 m/s
Dimensions: 2,4m * 1,1m * 1,1m	Dimensions: 6m * 1,5m * 2m
Number of operators: 1	Number of operators: 1
Price: $ 42870,00	Price: $ 50000,00

These machines are highly automated and only need one operator per machine to oversee that the machine works smoothly as well as to input new raw material. These machines also save a lot of labour costs.

Calculations

To determine the total kilometres that have to be produced by the drawing process, the length of a coil after being drawn to the different diameters was calculated by equating the volume of the coil before drawing and after drawing. In the equation below $r_i$and L_i are the radius of the raw material coils and length of raw material coil, respectively. Whereas r_f and L_f are the radius and length of a coil after the drawing process.

$$\pi r_i^2 L_i = \pi r_f^2 L_f$$

The weight per meter of the different thickness of wire needed is then calculated by dividing the weight of a coil (2.5 ton) by the length of the coil after being drawn. The total metres needed of drawn wire to produce one roll is calculated by weight per roll by the weight per meter. The amount of rolls needed per annum is then calculated by dividing the demand per annum by the weight per roll. The amount of rolls needed per annum is then multiplied by the meters per roll needed and then divided by 220 (the amount of working days per annum) to get the kilometres of drawn wire needed per day. A scrap allowance of 4% is then also added to compensate for expected scrapping of materials during this process. The results (rounded) from these equations can be seen in table 6.2.1 below.

Table 6.2.1: Drawing Process Calculations Results

Process	Bird Netting	Chicken Netting	Jackal Netting	Pig Netting	Welded Mesh	Totals
Wire radius (m)	0.0003	0.00045	0.0009	0.0009	0.00125	**-**
Length of coil after drawing (m)	486111	216049	54012	54012	28000	**838185**
Weight per meter (kg)	0.0051	0.0116	0.0463	0.0463	0.0893	**-**
Metres needed per roll	4375	3621	421	318	784	**-**
Rolls needed per annum	177778	131265	128205	544217	85714	**-**
Drawn wire needed per annum (km)	777778	475309	54012	172840	67200	**1547138**
Drawn wire needed per day with 4% scrap allowance (km)	3683	2251	256	818	318	**7325**

It is decided to not run the machines at full speed in order to decrease the wear and tear on the machines and the wires. The drawing speeds used can be seen in table 6.2.3 below. Detailed calculations and unrounded values of all calculations can be seen in Appendix A.

From here the number of drawing machines is calculated by taking the total kilometres of wire needed after drawing and dividing it by the total kilometres a drawing machine can draw in one working day. It is important to note that for the bird and chicken netting the wire goes through two drawing machines. One machine is used to take the wire diameter from 5mm to 1mm and another is used to further reduce the diameter to 0.6mm and 0.9mm as needed. From these calculations, it is evident that 36 drawing machines is needed to keep up with demand. A summary of the calculation results can be seen below in table 6.2.3.

Table 6.2.3: Drawing Machine Calculations

Process	Bird Netting	Chicken Netting	Jackal Netting	Pig Netting	Welded Mesh	Totals
Drawn wire needed per day with 4% scrap allowance (km)	3683	2251	256	818	318	**7325**
Machine 1 drawing speed (m/s)	15	15	20	20	20	**-**
Machine 2 drawing speed (m/s)	15	15	-	-	-	**-**
Length one machine draws (m)	378000	378000	504000	504000	504000	**2268000**
No. of machines needed	20	12	1	2	1	**36**

Workstation Layout

The workstation layout for the different drawing machines can be seen below in figures 6.2.4 and 6.2.5. Figure 6.2.4 indicates the layout for the jackal netting, pig netting and welded mesh. Figure 6.2.5 show the layout for the bird and chicken netting where two machines are needed to draw the desired diameter. A 1m walkway is indicated in yellow all around the machine for use by operators. All machines pull the wire to be drawn directly from the raw material storage.

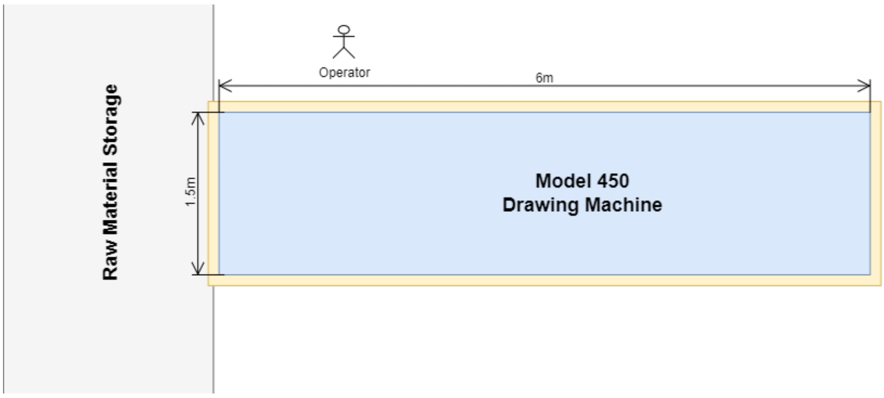

Figure 6.2.4: Welded Mesh and Jackal & Pig Netting layout

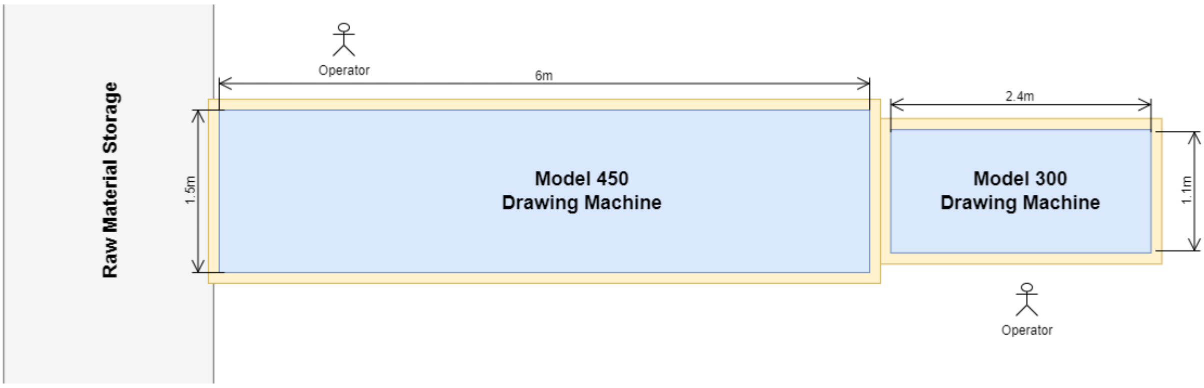

Figure 6.2.5: Bird & Chicken Netting layout

6.3. Operation B (Galvanising)

Galvanising

Specifications

Figure 6.3.1: Wire coiling and de-coiling

Figure 6.3.2: Zinc Ingot input section

Figure 6.3.3: Hot-dip galvanisation

Figure 6.3.4: Wire cooling

Table 6.3.1: Galvaniser Specifications

Dimensions (l*w*h)	105*5*3 (m)
Operators	Four
Throughput	600 kg/h
Voltage	380V 3P 50Hz
Price	R700 000
Zinc Coating Thickness	50 microns

Calculations

To determine the number of galvanisers needed, the hourly weight requirement of galvanised wire is calculated and divided by the throughput rate of 600kg/h (Alibaba, 2024). A rounded summary of these calculations is seen in table 6.3.2 below. Detailed calculations and unrounded values of all calculations can be seen in Appendix A.

Table 6.3.2: Galvaniser Calculations

Product	Annual demand (kg)	Hourly requirement (kg)	Machines needed
Bird Netting	4 000 000	2 598	4
Chicken Netting	5 500 000	3 572	6
Jackal Netting	2 500 000	1 623	3
Pig Netting	8 000 000	5 194	9
Welded Mesh	6 000 000	3 896	7
Total	**26 000 000**	**16 883**	**29**

Zinc Ingot Storage

Specifications

Zinc ingot slabs of 25kg each are used for the galvanisers (Boliden, 2022). The slabs are delivered and stored in pallets that consist of 10 layers with 8 zinc ingot slabs on each layer as can be seen in figure 6.3.5 below. One work week's (5 days) storage is needed for the zinc ingots.

Figure 6.3.5: Zinc Ingot Pallet

Calculations

To calculate the total weight of Zinc-ingots used daily, the total surface area of all wire that must be coated is divided by the galvanised coating weight per square meter. The rounded calculation results can be seen in table 6.3.3 below.

Detailed calculations and unrounded values of all calculations can be seen in Appendix A.

Table 6.3.3: Zinc Ingot Calculations

Product	Daily demand (m)	Wire Radius (m)	Surface Area (m^2)	Galvanised Coating (kg/m^2)	Zinc needed daily (kg)
Bird Netting	3682660	0.0003	6942	0.5	3471
Chicken Netting	2250514	0.00045	6363	0.5	3182
Jackal Netting	255740	0.0009	1446	0.5	723
Pig Netting	818369	0.0009	4628	0.5	2314
Welded Mesh	318182	0.00125	2499	0.5	1249
Total	**7325465**		**21878**		**10939**

From here it is calculated that 438 zinc ingots are needed daily and 2188 are needed for one work week. Using the dimensions of one pallet (1m*0.5m*0.4m) it is calculated that 14 m^2 of storage area is needed per galvaniser for one week.

Workstation Layout

The workstation layout for the galvanisers and zinc storage can be seen below in figure 6.3.6. A 1m walkway is indicated in yellow all around the machine for use by operators. It is also important to note that the zinc ingot storage is large enough so that it will also be used by the galvaniser put on the other side of the storage. There will thus be two galvanisers for each zinc ingot storage area.

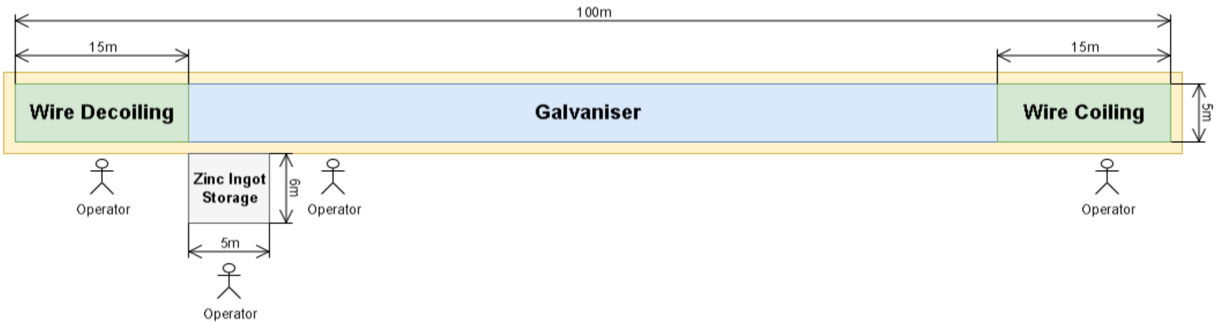

Figure 6.3.6: Galvaniser & Zinc Ingot storage layout

6.4. Operation C (Netting)

Specifications for Hexagonal Wire Netting Machine

Figure 6.4.1: Netting Machine Input

Figure 6.4.2: Netting Machine Output

Figure 6.4.1 above shows where the drawn and galvanized steel coils will be placed as input to the hexagonal wire netting machine. Figure 6.4.2 shows the output that the hexagonal wire netting machine produces.

Table 6.4.1: Hexagonal Wire Netting Machine Specifications

Description	Value
Max Netting (mm) Width:	4500
Max Diameter of Weaving Wire (mm)	2.7
Production Capacity (Sq. Meter / hour)	500
Dimensions (L*W*H)	32*15*6
Number of Operators	4

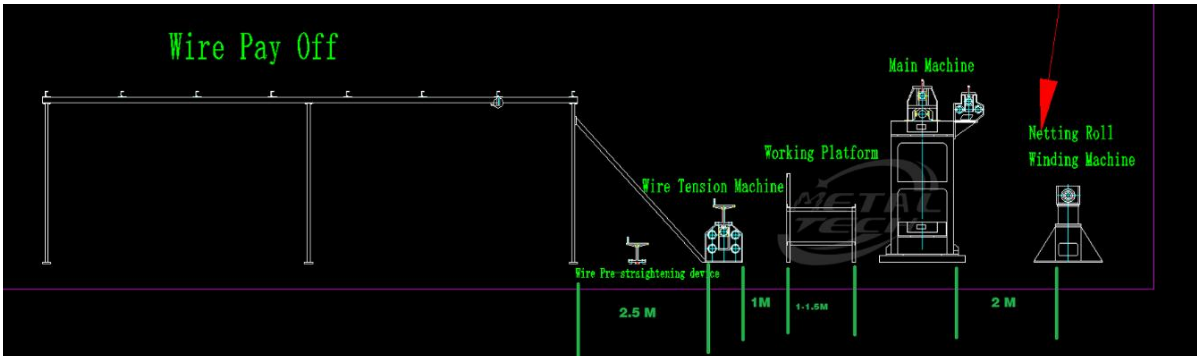

Figure 6.4.3: Netting process flow

The hexagonal wire netting machine's settings can be adjusted to the required aperture sizes by making adjustments to the twisting and spacing mechanisms of the machine.

Calculations for Hexagonal Wire Netting Machine

To determine how many hexagonal wire netting machines will be required to meet demand, we first had to determine how many rolls of each product were going to have to be produced per day.

Table 6.4.2: Netting Calculation Results

	Roll Length (m)	Roll Hight (m)	Roll Weight (kg)	Customer demand (Tons per Year)	Customer Demand (kg per Year)	Rolls per Year	Rolls per Day (Demand)
Bird Netting	50	0,9	22,5	4000	4000000	177778	808
Chicken Netting	50	1,8	41,9	5500	5500000	131265	597
Jackal Netting	50	0,6	19,5	2500	2500000	128205	583
Pig Netting	30	0,9	14,7	8000	8000000	544218	2474
Welded Mesh	30	2	70	6000	6000000	85714	390

From table 6.4.2 above we see that the weight of each product roll was taken and divided by the total annual demand in weight for each product to obtain the number of rolls that had to be produced per year. Then the number of annual rolls was further divided by the number of operational days in a year (220 days) and we then obtained the number of rolls that had to be produced per day for each product.

After doing research on the hexagonal netting machine's specifications, we found that this machine can net at a maximum rate of 500 square meter per hour (as indicated above). We don't however want to use this machine at its full capacity; hence we made the decision to have it operate at 450 square meter per hour.

Table 6.4.3 Netting Machine Calculations:

	Rolls per Day	Area per Roll (m^2)	Total Net Area per Day Needed (m^2)	Machine Production Rate (m^2 / h)	Machine Production Rate (m^2 / day)	Number of Netting Machines Needed
Bird Netting	808	45	36364	450	3600,00	10
Chicken Netting	597	90	53699	450	3600,00	15
Jackal Netting	583	30	17483	450	3600,00	5
Pig Netting	2474	27	66790	450	3600,00	19

From table 6.4.3 above we see that the area of a roll of each respective product was obtained by multiplying the roll height (that was take as the width) with the length of the roll. After obtaining the area of each roll, we calculated the required area to be produced per day to satisfy demand. We thus, multiplied rolls per day required with the area per roll for each respective product. Now that we have the total area to be netted per day for each product, we can now take the machines production rate of 450 square meter per hour to determine the production rate in square meters per day. The total netting area needed per day is then divided by the production rate per day to determine the number of netting machines that is needed in the new facility.

Specifications for Automatic Wire Mesh Welding Machine

Figure 6.4.4: Galvanised coils fed into Netting Machines

Figure 6.4.5: Automatic Wire Mesh Welding Machine Output

Table 6.4.4: Automatic Wire Mesh Welding Machine Specifications

Description	Value
Production Rate (times/min)	50
Machine Dimensions (L*W*H)	4,5m * 2,3m * 1.8m
Max Width of Mesh (m)	2,1m
Number of Operators	3

For the Automatic Wire Mesh Welding Machine, the specification for the production rate is specified slightly different. This machine's production rate was given in terms of times per minute, referring to the number of rows of wire that the machine welds per minute.

Calculations for Automatic Wire Welded Mesh Machine

Table 6.4.4: Automatic Wire Mesh Welding Machine Production Capacity

	Welded Mesh Production Rate (Times/min)	Aperture (m)	Production Rate (m/min)	Production Rate (m/h)	Production Rate (m^2 / h)	Welded Mesh Production Rate (m^2 / day)
Welded Mesh	50	0,05	2,5	150	300	2400

From table 6.4.4 we see that the number of times per minute was multiplied with the aperture to obtain the length of the roll that the machine can produce in meters per minute. It was then converted to meters per hour. This length was multiplied with the height of the roll (2m) to obtain the total area in square meters per hour that the automatic wire mesh welding machine can produce. This production rate was then converted to square meters per day assuming an 8-hour workday.

Table 6.4.5: Number of Automatic Wire Mesh Welding Machines needed

	Rolls per Day	Area per Roll (m^2)	Total Net Area per Day Needed (m^2)	Machine Production Rate (m^2 / h)	Machine Production Rate (m^2 / day)	Number of Wire Mesh Welding Machines Needed
Welded Mesh	390	60	23377	300	2400	10

From table 6.4.5 above we see that total number of rolls required per day for welded mesh was multiplied with the area per roll to obtain the total net area per day needed. This total was then divided by the machine production rate in square meters per day to obtain the number of automatic wire mesh welding machines that will be required to meet demand.

Workstation Layouts for Hexagonal Wire Netting Machine

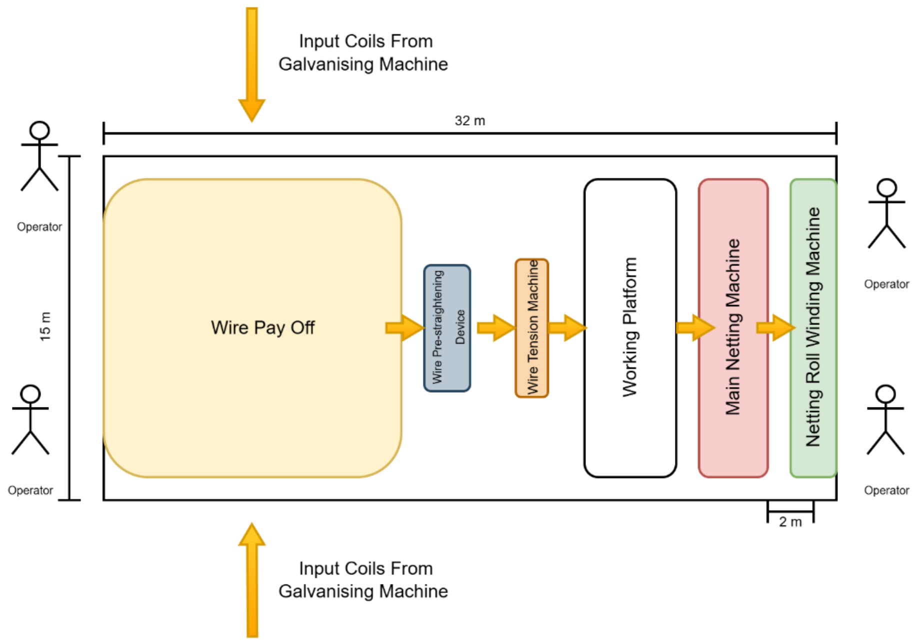

Figure 6.4.6: Hexagonal Wire Netting Machine Workstation Layout

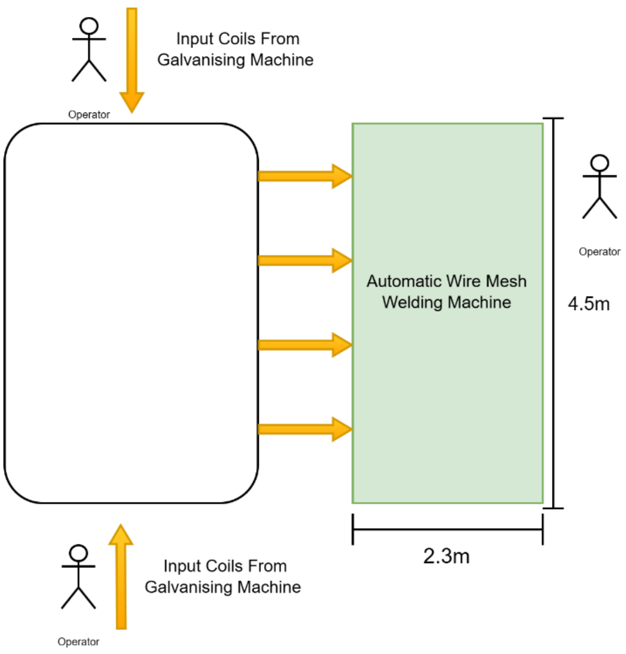

Figure 6.4.7: Automatic Wire Mesh Welding Machine Workstation Layout

6.5. Operation D (Packaging)

Specifications

The packaging of the finished goods will be the same for all the products; Bird netting, chicken netting, Jackal netting, pig netting and welded mash. The finished products are packaged using a wire mash roll packing machine, that uses a High-Density Polyethylene (HDPE). The plastic protects the products from getting damaged during storage and transportation. Every machine requires 1 operator. After wrapping the wire mesh, it is stored on Pallets. A single pallet carries a maximum of 1000 kg. This is made to accommodate customers with small orders that can carry their products on the back of their vans. The image below depicts the packing machines used for packaging.

Figure 6.5.1: Packaging Machine

The image below shows the dimensions of the packaging machine and each machine requires one operator.

Figure 6.5.2: Dimensions of Packaging Machine

Calculations

The packaging process is a critical step in ensuring that the finished netting products are properly wrapped and prepared for distribution. The table below outlines key operational metrics, including the number of rolls processed per day, packaging efficiency, required packaging machines, and the space and personnel needed for each product type. With a packaging efficiency of 50 rolls per hour, the facility operates multiple machines to meet daily production demands. The total number of packaging machines required is 97, occupying a combined space of 434.63 square meters. Additionally, 97 personnel are allocated to operate the machines and manage the packaging workflow. This data provides a basis for optimizing packaging operations to improve efficiency and reduce space utilization.

Table 6.5.1: Space Requirements for Packaging

	Rolls per Day	Packaging Efficiency (Rolls/hour)	Number of Packaging Machine	Machine Dimension (square meters)	Number of Personnel	Total Space (square meters)
Bird Netting	808,08	50	16	4,48	16	72,40
Chicken Netting	596,66	50	12	4,48	12	53,46
Jackal Netting	582,75	50	12	4,48	12	52,21
Pig Netting	2473,72	50	49	4,48	49	221,65
Welded Mesh	389,61	50	8	4,48	8	34,91
Totals			**97**		**97**	**434,63**

Workstation Layout

The workstations of the packaging area are organized in a group cell format. Where all the machines working on the chicken netting are grouped together and similarly with the other machines for other types of products. This is done to increase the efficiency of each product. The image below depicts the layout of the packaging area.

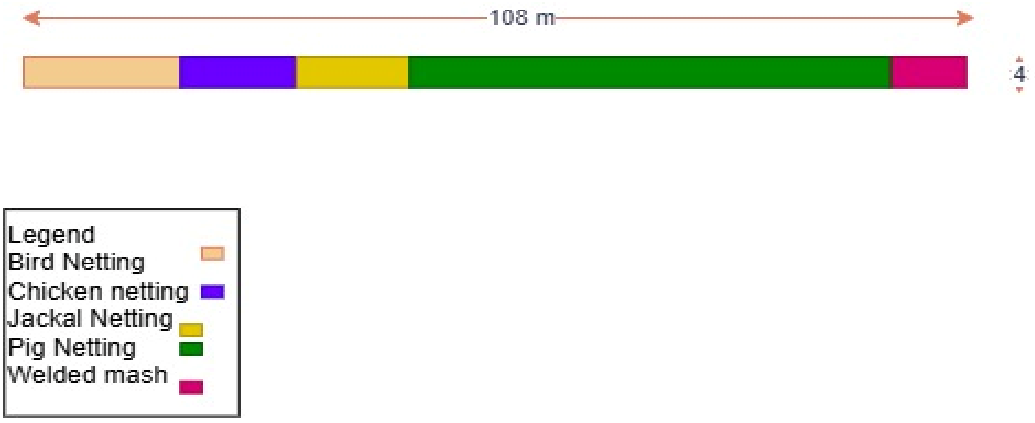

Figure 6.5.3: Packaging Workstation Layout

6.6. Finished goods storage & distribution

Small Orders

Small orders are defined of orders of 1 ton or less and makes up 10% of total production. A week's storage space is needed for small orders. Assuming that rolls are stacked on each other in pallets of 20 (four layers of four rolls each), the weekly storage space calculations can be seen in the table below. A week is also assumed to be 7 days.

Table 6.6.1: Small Order Storage Area Calculations

Product	Rolls produced per day	Total daily area needed for stacks of rolls (m^2)	Weekly area needed for storage (m^2)
Bird Netting	81	20.25	141.75
Chicken Netting	60	15	105
Jackal Netting	59	14.75	103.25
Pig Netting	248	62	434
Welded Mesh	39	9.75	68.25
Totals	**487**	**121.75**	**852.25**

Large Orders

The same calculations and assumptions as above can now be used for the large orders storage area. There however only needs to be enough space for two days storage.

Table 6.6.2: Large Order Storage Area Calculations

Product	Rolls produced per day	Total daily area needed for stacks of rolls (m^2)	Total area needed for storage (m^2)
Bird Netting	728	182	364
Chicken Netting	537	134.25	268.5
Jackal Netting	525	131.25	262.5
Pig Netting	2227	556.75	1113.5
Welded Mesh	351	87.75	175.5
Totals	**4366**	**1092**	**2184**

Delivery Trucks

The trucks that are we are going to use to deliver the finished product is a Volvo FH-16, which can carry a capacity of 70 tons in one load. The image below depicts the truck that is used.

Figure 6.6.1: Delivery Truck

7. Management Policies

Manufacturing wire mesh products such as chicken netting, pig netting, jackal netting, and welded wire mesh requires strict adherence to safety protocols and factory standards to ensure both worker safety and product quality. The factory layout must accommodate proper safety measures, including workroom doors that are at least 1.2 meters wide and 2 meters high. Floors should be covered with rubber sheets no less than 3 millimeters thick to prevent slips and electrical hazards. Maintaining cleanliness in work areas is essential, with strict housekeeping practices to keep pathways clear and address material spillages promptly.

To prevent workplace accidents, machine guarding is critical. Protective guards must be installed on all moving machinery to shield operators from potential injuries. Electrical safety is another priority, requiring the use of flameproof conduits and the avoidance of non-conductive handle lamps to minimize electrical hazards. Additionally, workers must be equipped with appropriate personal protective equipment (PPE), including gloves, aprons, and breathing apparatuses, particularly in areas where exposure to hazardous materials or fumes is a concern.

By implementing these safety protocols and adhering to recognized industry standards, factories can ensure efficient and secure production environments while delivering high-quality wire mesh products.

8. Sustainable Energy Solutions

8.1. Solar Power with Battery Storage

Solar energy is one of the most practical and sustainable solutions. By installing solar panels on the facility's roof or surrounding land, the facility can generate a significant portion of its electricity needs from renewable sources. Steel netting production involves energy-intensive processes such as wire drawing, hot-dip galvanising and cutting, making energy costs a major operational factor. Solar power can offset these costs by providing a steady supply of electricity during daylight hours. Additionally, integrating battery storage ensures that excess solar energy generated during the day can be stored and used during peak demand periods or when it is loadshedding, thus increasing energy reliability. Beyond cost savings, solar power reduces dependence on the conventional grid, making the facility more resilient fluctuations in electricity prices.

8.2. On-Site Wind Energy (Small-Scale or Hybrid System)

Wind energy presents another viable option, particularly if the facility is located in a region with sufficient wind speeds. Small to medium-scale wind turbines can be installed on-site to generate electricity, either as a standalone system or in combination with solar power for a hybrid energy solution. Wind energy is particularly advantageous because it can produce power at night and during overcast conditions when solar panels may be less effective. Vertical-axis wind turbines (VAWTs) can be considered if space is a constraint, as they are compact and well-suited for industrial environments. Once installed, wind turbines have low operational costs and provide long-term energy savings.

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Appendix A (Calculations)

Drawing

Product	Height of roll (m)	Aperture (mm)	Wire Diameter (m)	Roll length (m)	Weight per roll (kg)	Galvinized finish	Demand per annum (tons)
Bird Netting	0,9	13	0,0006	50	22,5	Light	4000
Chicken Netting	1,8	25	0,0009	50	41,9	Light	5500
Jackal Netting	0,6	75	0,0018	50	19,5	Light	2500
Pig Netting	0,9	90	0,0018	30	14,7	Light	8000
Welded Mesh	2	50	0,0025	30	70	Heavy	6000
Totals							26000

Weight of steel wire coil	2500		
Length of raw material coil (m)	7000	Original diameter (m)	0,005

	Length of coil on exit (m)	Weight per meter (kg)	Metres per roll	Rolls per annum	m needed per annum	km needed per annum
Bird Netting	486111,11	0,0051	4375,00	177777,78	777777777,8	777777,7778
Chicken Netting	216049,38	0,0116	3620,99	131264,92	475308642	475308,642
Jackal Netting	54012,35	0,0463	421,30	128205,13	54012345,68	54012,34568
Pig Netting	54012,35	0,0463	317,59	544217,69	172839506,2	172839,5062
Welded Mesh	28000,00	0,0893	784,00	85714,29	67200000	67200
Total	838185,19	0,1986	9518,88	1067179,80	1547138271,60	1547138,27

Days worked per day 220

	After draw		Before draw		With 4% scrap allowance (after draw)		With 4% scrap allowance (before draw)	
	m per day	km per day	m per day	km per day	m per day	km per day	m per day	km per day
Bird Netting	3535353,535	3535,353535	50909,09091	50,90909091	3682659,933	3682,659933	53030,30303	53,03030303
Chicken Netting	2160493,827	2160,493827	70000	70	2250514,403	2250,514403	72916,66667	72,91666667
Jackal Netting	245510,6622	245,5106622	31818,18182	31,81818182	255740,2731	255,7402731	33143,93939	33,14393939
Pig Netting	785634,119	785,634119	101818,1818	101,8181818	818368,8739	818,3688739	106060,6061	106,0606061
Welded Mesh	305454,5455	305,4545455	76363,63636	76,36363636	318181,8182	318,1818182	79545,45455	79,54545455
Total	7032446,689	7032,446689	330909,0909	330,9090909	7325465,301	7325,465301	344696,9697	344,6969697

	Daily Production Requirem	Drawing Speed (m/s)	Drawing Speed machine 2	Length one machine draws per day	No. of machines needed
Bird Netting	3682659,933	15	15	378000	20
Chicken Netting	2250514,403	15	15	378000	12
Jackal Netting	255740,2731	20	-	504000	1
Pig Netting	818368,8739	20	-	504000	2
Welded Mesh	318181,8182	20	-	504000	1
Total	7325465,301	90		2268000	36
				*use for 7hrs per day (1hr cleaning and setup time)	

No. of raw material coils needed	49,24242424
No. of raw material coils in storage (3 days)	147,7272727

	Height of roll (m)	Roll length (m)	Material thickness (m)
Bird Netting	0,9	50	0,0008
Chicken Netting	1,8	50	0,0011
Jackal Netting	0,6	50	0,002
Pig Netting	0,9	30	0,002
Welded Mesh	2	30	0,0027

Galvanisation

Avg galvinisation coating thickness (µm)	50		
Avg galvinisation coating thickness (m)	0,00005		
Total metal to be galvinised per day (km)	7032,446689		
Zinc density (g/cm^3)	7,14	pi	3,141592654
Zinc mass per unit area (g/m^2)	357		

	Wire diameter (m)	Surface area per meter (m^2)	Zinc mass per meter (g)	Zinc mass per km (kg)
Bird Netting	0,0006	0,00204	520,5106947	520,5106947
Chicken Netting	0,0009	0,00298	760,7464	760,7464
Jackal Netting	0,0018	0,00581	1481,453516	1481,453516
Pig Netting	0,0018	0,00581	1481,453516	1481,453516
Welded Mesh	0,0025	0,00801	2042,003495	2042,003495
Total				**6286,16762**

Galvaniser dimensions:	
Length (m)	100
Width (m)	5
Depth (m)	3
Galvanising speed (kg/h)	600

	yearly requirement (kg)	daily requirement (kg)	hourly requirement (kg)	machines needed	
Bird Netting	4000000	18181,81818	2597,402597	4,329004329	*30 mins needed each morning to replenish galvanisers
Chicken Netting	5500000	25000	3571,428571	5,952380952	*work with 7hrs per day of galvanising time (1hr setup and cleaning time)
Jackal Netting	2500000	11363,63636	1623,376623	2,705627706	
Pig Netting	8000000	36363,63636	5194,805195	8,658008658	
Welded Mesh	6000000	27272,72727	3896,103896	6,493506494	
	26000000	118181,8182	16883,11688	29	

Zinc ingot weight (kg)	25				
	Daily demand (m)	Wire radius (m)	Surface area(M^2)	gavlanised coating (kg/m^2)	daily zinc demand (kg
Bird Netting	3682659,933	0,0003	6941,650434	0,5	3470,825217
Chicken Netting	2250514,403	0,00045	6363,179565	0,5	3181,589782
Jackal Netting	255740,2731	0,0009	1446,177174	0,5	723,0885869
Pig Netting	818368,8739	0,0009	4627,766956	0,5	2313,883478
Welded Mesh	318181,8182	0,00125	2498,994156	0,5	1249,497078
	7325465,301		21877,76828		**10938,88414**

No. of ingots needed daily	437,55537
No. of ingots needed for work week	2187,776828
Length of pallet (m)	1
Width of pallet (m)	0,5
Height of pallet (m)	0,4
No. of layers in pallet	10
No. of slabs on a pallet layer	8
Area for 5 days storage (sqm)	13,67360518

Appendix B (Minutes of Meetings)

Minutes of Meeting (3 March 2025)

Time:	10:00-11:00
Location:	Engineering Campus Mini-SS
Meeting called to order by:	Karla van Dyk
Minutes of meeting taken by:	Karla van Dyk

Attendance:

Sheldon Farmer

Karla van Dyk

Absent:	**Reason:**
Ofentse Mathebula	Did not have transport to the meeting

Agenda:

1. Group member introduction
2. Talk-through of assignment
3. Initial job allocation

Decisions:

1. Initial work duties (due on 11 March 2025) were divided as follows:

Person Responsible	Duties
Sheldon Farmer	• Research wire drawing machines • Research wire braiding machines • Calculations about demand • Workstation layouts of drawing and netting process
Ofentse Mathebula	• Research packaging and packaging process • Workstation layout for packaging process • Research safety and factory requirements • Research delivery truck capacities
Karla van Dyk	• Research sustainable energy options • Research galvanising process • Calculations about galvanising process • Calculations about raw material

2. All members agreed to attend the *class session on 4 March 2025 to work on assignment and ask any questions that might arise.

**Mr Mathebula did not end up attending the class session.*

3. Ms Van Dyk will create a document for everyone to edit on and will also format this document. (link:https://docs.google.com/document/d/1atGOHg7HVeIZ1RXGH4uANZLSe_n94bHxI8xfl-fTWPY/edit?usp=sharing)
4. If it were to rain during future meetings, either Mr Farmer or Ms Van Dyk will provide transport for Mr Mathebula.

Next Meeting:

The next meeting is scheduled for Wednesday 12 March 2025 at 11:00 at the Main Campus SS. Notice for this meeting was sent out on Friday 7 March 2025 via the Whatsapp group.

Minutes of Meeting (12 March 2025)

Time:	11:00-12:00
Location:	Main campus SS
Meeting called to order by:	Sheldon Farmer
Minutes of meeting taken by:	Karla van Dyk

Attendance:

Sheldon Farmer

Ofentse Mathebula

Karla van Dyk

Agenda:

1. Check-in on progress of group members
2. Next phase of work allocation
3. Small orders

Decisions:

1. Work duties (due on 19 March 2025) were divided as follows:

Person Responsible	Duties
Sheldon Farmer	• Initial workstation layout for drawing process • Initial workstation layouts of netting process
Ofentse Mathebula	• Packaging calculations • Initial workstation layout for packaging process
Karla van Dyk	• Research sustainable energy options • Initial workstation layout for galvanising • High-level process flow drawing

2. A small order is defined as an order that can be picked up by customers themselves (10 rolls or less) and it constitutes 10% of total production.
3. Start writing report after next week's meeting

Next Meeting:

Next meeting is scheduled for Wednesday 19 March 2025 at 11:00 at the Main Campus SS.

Minutes of Meeting (19 March 2025)

Time:	11:00-12:00
Location:	Main campus SS
Meeting called to order by:	Karla van Dyk
Minutes of meeting taken by:	Karla van Dyk

Attendance:

Sheldon Farmer
Ofentse Mathebula
Karla van Dyk

Agenda:

1. Check-in on progress of group members
2. Assess what needs to be done before submission
3. Internal submission deadline
4. Material handling

Decisions:

1. All members progressed well with their responsibilities from the previous week

2. Work duties (due on 23 March 2025) were divided as follows:

Person Responsible	Duties
Sheldon Farmer	• Write background for report • Complete workstation layout • Write detailed design specifications for drawing and netting
Ofentse Mathebula	• Write design objectives for report • Complete workstation layout • Complete calculations for packaging • Write detailed design specifications for packaging
Karla van Dyk	• Write assumptions and constraints for report • Complete workstation layouts • Write detailed design specifications for galvanising, raw material storage and finished good storage • Format report and label figures

3. Report must be completed by Sunday 23 March 2025
4. Need to research material handling specifically between the drawing machines and galvanisers. Particularly look at some sort of forklift.

Next Meeting:

Next meeting is scheduled for Wednesday 26 March 2025 at 11:00 at the Main Campus SS. This meeting will be used to discuss what still needs to be done for the final submission.